Scribe Question 1: Baseline System and Motivation for Randomization

The deterministic model implemented in milestone-3 is a rule-based traffic signal control mechanism in which the timings of the signal are decided using a fixed decision criterion. At every decision step, metrics such as queue length, waiting time, etc are evaluated for each lane and assigns the green signal to the lane with the highest value. This approach is purely deterministic, i.e., for a given congestion state the system will always produce the same decision. The cycle time is also then analytically determined at every step and there is no randomness involved in the decision-making process.

While this baseline model provides a predictable control strategy, its only limitation is to handle a real-world situation where the traffic condition is very stochastic. Traffic conditions are almost never uniform and can significantly vary over time due to driver behaviour, uneven traffic distribution, unknown reasons for congestion such as accidents, etc. Because of which, the deterministic approach fails almost every time and may lead to other side-effects such as lane starvation, where certain lanes are constantly deprioritized.

To address such issues, randomness has to be introduced into the system. This modification specifically targets the decision rule which is used for signal selection. Instead of choosing lanes with maximum queue length, the system has to consider a probabilistic mechanism that always allows for more flexibility and adaptability in decision-making. This change enables the system to respond to uncertainties in traffic conditions in a better way in turn, improving overall performance.

Scribe Question 2: Randomized Algorithm Design

The proposal randomized algorithm is based on the Las Vegas algorithm which uses probabilistic decision-making along with deterministic validation. First the algorithm identifies the lanes that exhibit high queue lengths compared to other lanes. Instead of selecting a lane with the highest queue length, the algorithm randomly selects one lane from the subset.

The randomness is thus introduced very specifically in the lane selection component, where the lane is selected using a probabilistic sampling mechanism. This can be understood as sampling from a uniform distribution over a set of candidate lanes, ensuring that every high-priority lane has a positive probability of being selected. This allows the model to explore multiple options instead of repeatedly favouring a single lane.

After a lane is selected, a validation step is performed to ensure that the choice made for the lane does not lead to a poor system performance. If the lane leads to an undesirable outcome or does not meet certain criteria then the algorithm goes back to the deterministic baseline decision rule. This hybrid decision makes sure that correctness is maintained while also benefitting from the flexibility which is introduced by the randomness in the selection process.

Scribe Question 3: Probabilistic Reasoning Behind the Algorithm

The motivation behind introducing a randomness into the algorithm is because of the probabilistic nature of traffic systems. The arrivals of vehicles are commonly modelled from Poisson distribution, reflecting the nature of the flow of traffic being very random.

A deterministic algorithm assumes that the level of predictability does not go hand in hand with these stochastic characteristics of a real-world traffic scenario. Due to which, it repeatedly makes decisions which are not optimum when faced with an uncertain traffic pattern. In comparison, a randomized algorithm considers a probabilistic reasoning in order to account for uncertainties. Because of such a randomness in the algorithm, it considers other strategies instead of committing to a single fixed strategy and also explore multiple action possibilities.

The hybrid Las Vegas algorithm strengthens this approach by combining randomness with validation ensuring that correctness is not sacrificed for flexibility. Along with that, this randomized strategy expects to improve the system's performance by reducing metrics such as waiting time, queue lengths and enhancing adaptability under uncertain traffic conditions. Such expectations are also supported by simulation results which demonstrate a significant improvement in the traffic conditions over the deterministic baseline.